

CSA 0307 - Data Structure

CO5-AT1, Pseudocode Writing

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B.Tech IT

SLOT A

1. Greedy Graph coloring for scheduling:

Greedy - coloring(G)

1. for each vertex v in G :

color $[v] \leftarrow 0$

2. For each vertex v in G :

Create an empty set used colors.

For each neighbor u of v :

If color $[u] \neq 0$:

Add color $[u]$ to used colors.

color $[v] \leftarrow 1$

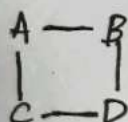
While color $[v]$ is in used:

color $[v] \leftarrow$ color $[v] + 1$

Return color $[v]$

// color $[v]$ represents the time slot of Task v .

eg:



2. Max - Min Path / Maximum Bottleneck capacity:

MAX-MIN-PATH (G, s, D)

1. For each vertex v in G :

capacity [v] $\leftarrow 0$

Parent [v] $\leftarrow \text{NIL}$

2. Capacity [s] $\leftarrow \infty$

3. Create a priority queue Q

Insert (s, ∞) into Q .

4. While Q is not empty:

$u \leftarrow$ vertex with Maximum

capacity from Q

If $u = D$

Break

for each neighbor v of u :

new capacity $\leftarrow$

$\text{MIN}(\text{capacity}[u], \text{Bandwidth}(u, v))$

If new capacity $>$

capacity [v]: in Q

5. Reconstruct the path from D to s using parent [$]$

6. Reverse the path

7. Return:

Path from s to D

Maximum bottleneck, capacity = capacity [D]

Example:

